System Design Document (SDD)

ServiceMate - Local Service Booking Platform

1. Introduction

ServiceMate is a web-based platform that connects users with nearby service providers such as electricians, plumbers, cleaners, and technicians.

The platform allows users to discover services, book professionals, and manage service appointments online.

The system is built using the MERN stack (MongoDB, Express.js, React.js, Node.js) with Vite as the frontend build tool for faster development and optimized performance.

2. System Objectives

• Provide a platform for users to find local services.

• Allow service providers to list and manage their services.

• Enable users to book services online.

• Maintain booking records and service history.

• Provide secure authentication for users and providers.

3. System Overview

ServiceMate follows a client-server architecture where the frontend communicates with the backend through REST APIs.

Main components:

• Frontend Application – React + Vite

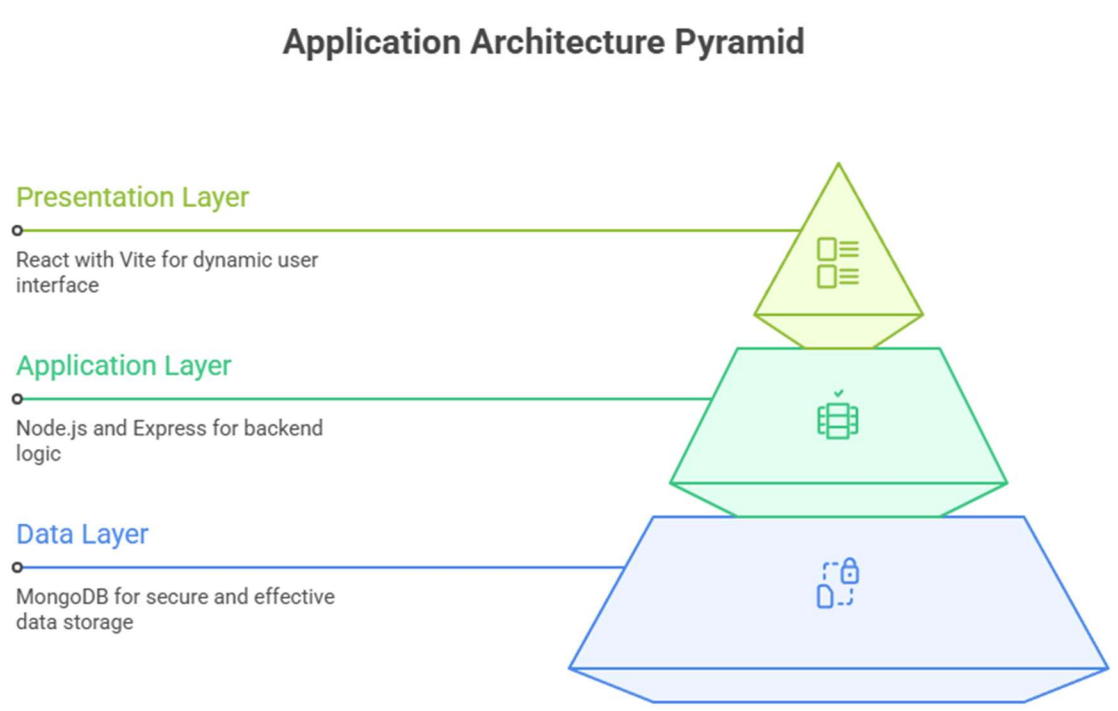
• Backend API – Node.js + Express

• Database – MongoDB

• Authentication system for login and signup

4. System Architecture

ServiceMate follows a three-layer architecture:



5. Technology Stack

Frontend:

• React.js

• Vite

• JavaScript (ES6)

• HTML5, CSS

Backend:

• Node.js

• Express.js

• REST APIs

Database:

• MongoDB

• Mongoose ODM

Development Tools:

• Postman

• Git & GitHub

• VS Code

6. Database Design

Main Collections:

• Users

• ServiceProviders

• Services

• Bookings

• Payments

• Reviews

7. Users Collection

Fields:

• _id

• name

• email

• password

• phone

• role

• createdAt

Roles:

• Customer

• Service Provider

• Admin

8. Services Collection

Fields:

• _id

• serviceName

• description

• price

• providerId

• category

9. Bookings Collection

Fields:

• _id

• userId

• serviceId

• providerId

• bookingDate

• status

• paymentStatus

10. Reviews Collection

Fields:

• _id

• userId

• providerId

• rating

• comment

• date

11. Database Relationships

• One User can create multiple Bookings.

• One Service Provider can offer multiple Services.

• One Service can have multiple Bookings.

• One Provider can have multiple Reviews.

12. API Design

Authentication APIs:

Method	Endpoint	Purpose
POST	/api/auth/signup	Register new user
POST	/api/auth/login	Login user

Service APIs:

Method	Endpoint	Purpose
Get	/api/services	Fetch services
POST	/api/services	Add service

Booking APIs:

Method	Endpoint	Purpose
POST	/api/bookings	Create booking
POST	/api/bookings	Get user bookings

13. Frontend Design

Main pages:

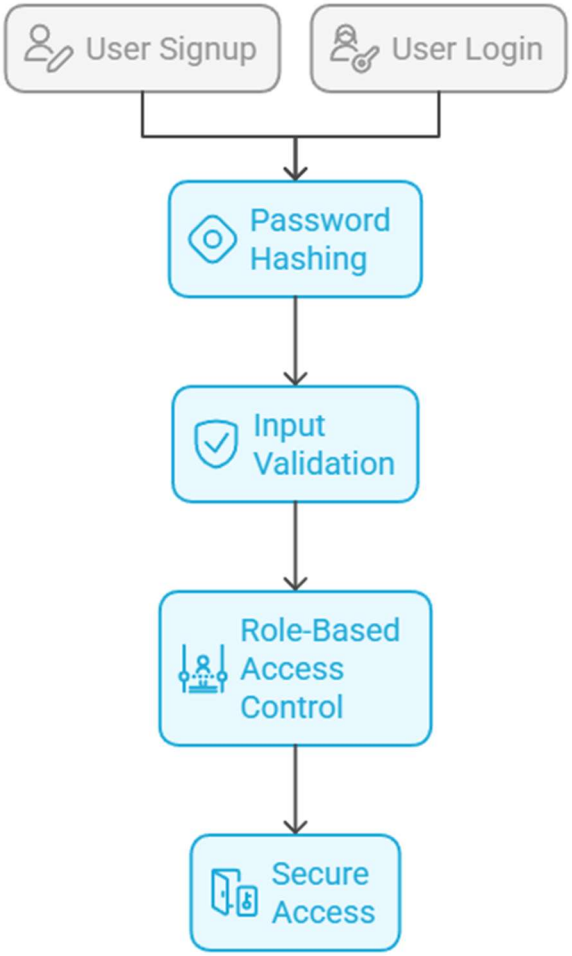
• Home Page

- Service Listing Page
- Service Details Page
- Booking Page
- User Dashboard
- Provider Dashboard
- Login Page
- Signup Page

React Project Structure:

```
src/
  components/
  pages/
  services/
  App.jsx
  main.jsx
```

14. Authentication Design



15. Deployment Architecture

Frontend	Vercel (React + Vite)
Backend	Render / Node Server
Database	MongoDB Atlas

16. Security Considerations

• Password hashing

• Input validation

• Error handling

• Secure API endpoints

• Role-based access control

17. Future Enhancements

• Online payment integration

• Real-time booking updates

• Provider ratings and reviews

• Mobile application

• Notification system

18. Conclusion

ServiceMate is designed using the MERN stack to provide a scalable and efficient service booking platform.

The modular architecture allows future feature expansion while maintaining performance and security.